Information

⚠️ **Security Warning**: Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

📋 Challenge Description

Files can always be changed in a secret way. Can you find the flag?

💡 Hints

- Look at the details of the file
- Make sure to submit the flag as picoCTF{XXXXX}

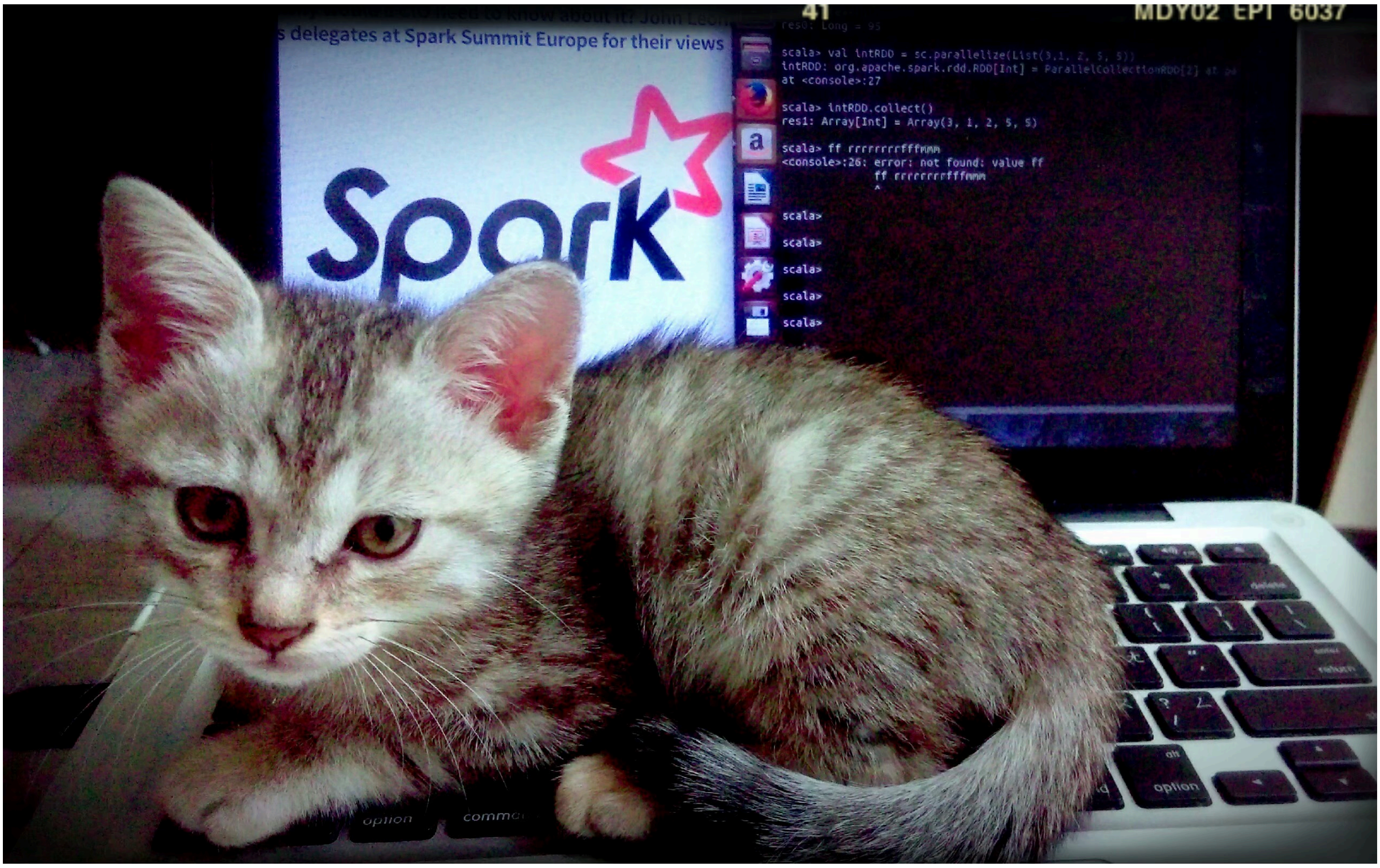
🧰 Tools Used

- **file**: Utility to identify file types
- **strings**: Command-line tool to extract readable ASCII sequences from files
- **grep**: Text search utility for filtering output
- **exiftool**: Metadata inspection tool
- **base64**: Encoding/decoding utility for Base64 data

🔍 Complete Solution

Step 1: Download and Prepare

Download the file named `cat.jpg` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file cat.jpg
```

Expected output:

```
cat.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density 1x1, segment
length 16, baseline, precision 8, 2560x1598, components 3
```

Step 3: Search for Visible Text

The flag may be present as plain text. Try extracting all character strings:

```
strings cat.jpg | grep -i "picoCTF"
```

Expected output:

```
PicoCTF
    <rdf:li xml:lang='x-default'>PicoCTF</rdf:li>
```

Step 4: Inspect Metadata

This is the crucial step! Use `exiftool` to examine all image metadata:

```
exiftool cat.jpg
```

Complete output:

```
ExifTool Version Number         : 12.76
File Name                       : cat.jpg
Directory                       : .
File Size                       : 878 kB
File Modification Date/Time     : 2026:01:04 12:47:00+01:00
File Access Date/Time           : 2026:05:08 17:11:44+01:00
File Inode Change Date/Time     : 2026:01:04 12:47:00+01:00
File Permissions                : -rwxrwxrwx
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.02
Resolution Unit                 : None
X Resolution                    : 1
Y Resolution                    : 1
Current IPTC Digest             : 7a78f3d9cfb1ce42ab5a3aa30573d617
Copyright Notice                : PicoCTF
Application Record Version      : 4
XMP Toolkit                     : Image::ExifTool 10.80
License                         : cGljb0NURnt0aGVfbTN0YWRhdGFfMXNfbW9kaWZpZWR9
Rights                          : PicoCTF
Image Width                     : 2560
Image Height                    : 1598
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 2560x1598
Megapixels                      : 4.1
```

🎯 **Key point**: Notice the `License` field which contains a suspicious, encoded string!

Step 5: Decode the Base64 Value

The string in the `License` field is encoded in Base64:

```
cGljb0NURnt0aGVfbTN0YWRhdGFfMXNfbW9kaWZpZWR9=
```


Decode it using one of these commands:

```
echo 'cGljb0NURnt0aGVfbTN0YWRhdGFfMXNfbW9kaWZpZWR9' | base64 -d
```

✅ Final Result

```
picoCTF{<REDACTED>}
```

🧠 Key Concepts

- **Metadata**: Hidden information in files
- **Base64**: Common encoding to obscure or compress content
- **Anomaly Recognition**: Looking for elements that seem suspicious or out of place